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UNNI C02 SMARTIFICATION – REV A

Two-sheet cleanup: standard power symbols, grid-aligned wiring, explicit local connectivity.

USB + Power

ESP32–C6 + Auxiliary

Simulation Harness

File: 01_usb_power.kicad_sch

File: 02_esp_aux.kicad_sch

File: 03_simulation_harness.kicad_sch

Unni C02 Smartification

Sheet: /
File: unni-smartification-c6.kicad_sch

Title: Unni C02 Smartification – ESP32–C6 Rev A

Size: A3Date: 2026–08–19

KiCad E.D.A. 10.99.0–2795–g77a613d64cId: 1/4

A

B

C

D

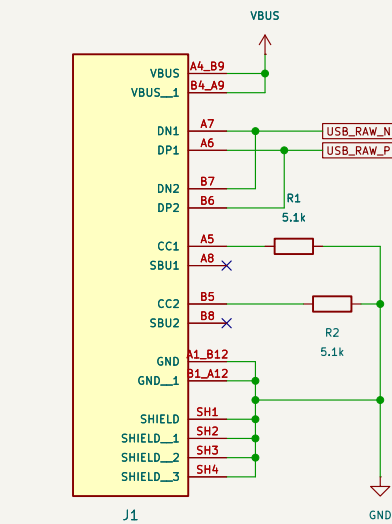
E

F

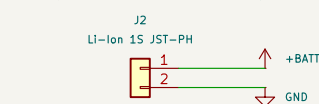
12345678

USB + POWER

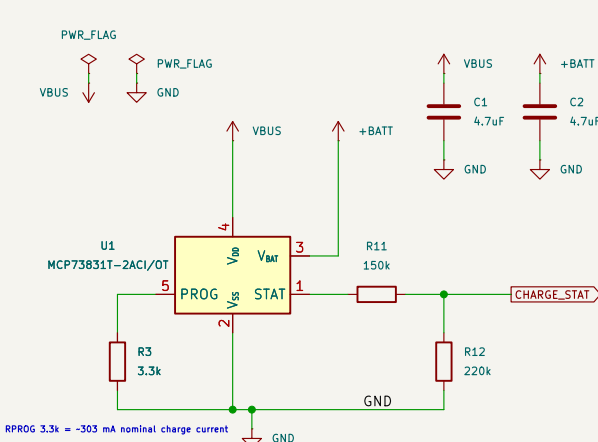
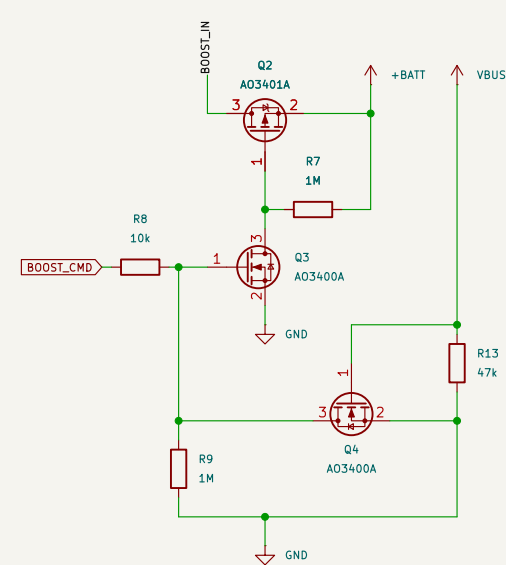
All placement/wiring is on 50 mil (1.27 mm) grid. Standard KiCad symbols used where available.



J1
USB-C (replacement daughterboard)

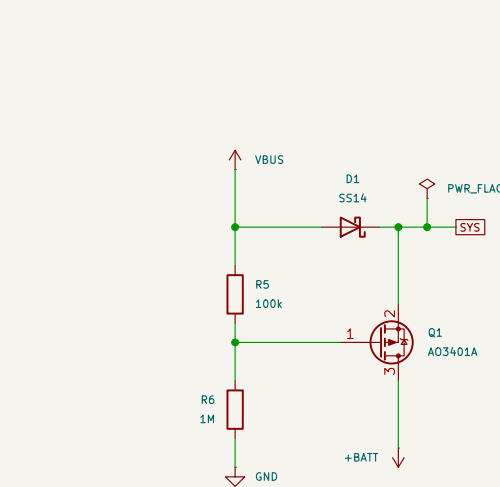


FORCED-AWAKE 5V

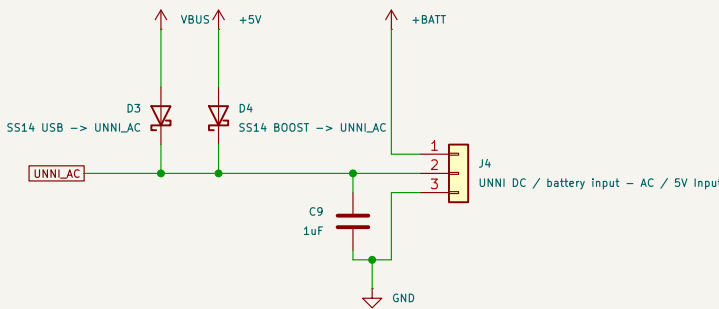
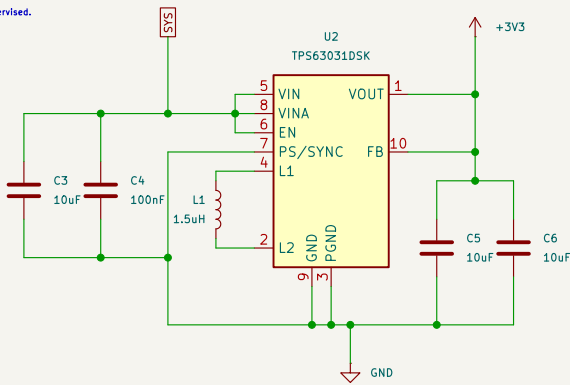
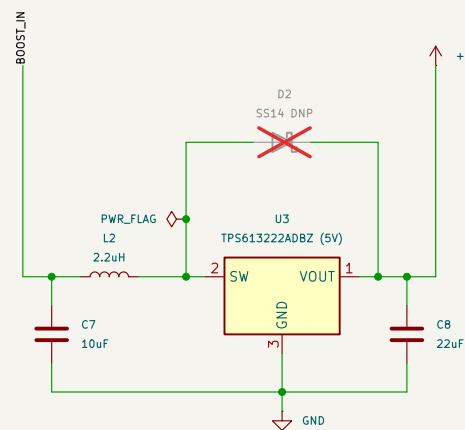


RPROG 3.3k = -303 mA nominal charge current

MCP73831 runs autonomously: R3 = 3.3 kΩ sets ~300 mA; STAT is monitored by the ESP. Battery NTC remains firmware-supervised.



UNNI POWER OUTPUTS



Unni C02 Smartification

Sheet: /USB + Power/
File: 01_usb_power.kicad_sch

Title: USB + Power

Size: A3
KiCad E.D.A. 10.99.0-2795-g77a613d64c

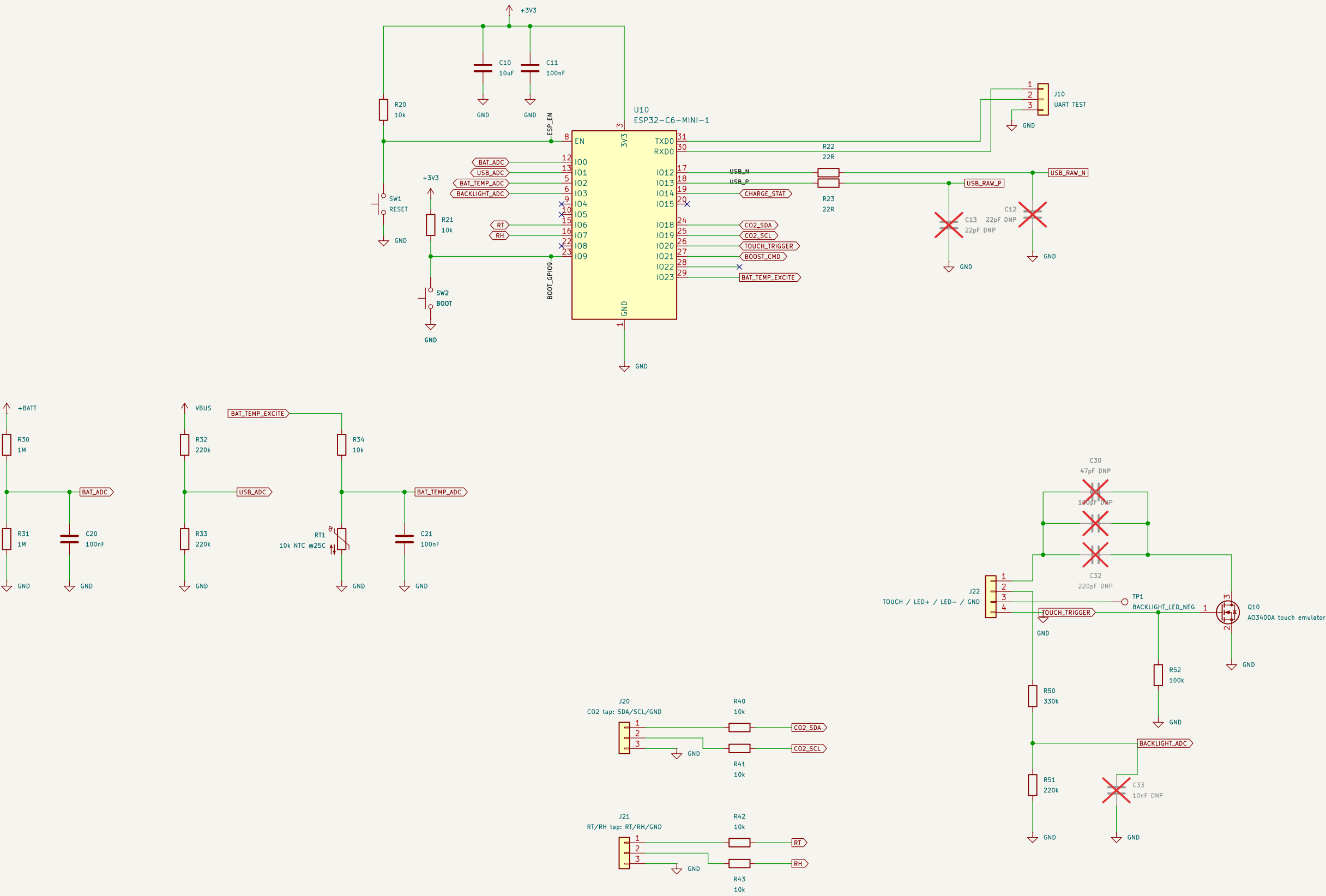
Date: 2026-08-19

Rev: A-v29-validated

Id: 2/4

ESP32-C6 + AUXILIARY CIRCUITS

GPIO0-3 are reserved for ADC inputs; RT/RH use RTC-capable GPIO6/7 for deep-sleep wake.



Unni CO2 Smartification

Sheet: /ESP32-C6 + Auxiliary/
File: 02_esp_aux.kicad_sch

Title: ESP32-C6 + Auxiliary Circuits

Size: A3 Date: 2026-08-19 Rev: A-v29-validated
KiCad E.D.A. 10.99.0-2795-g77a613d64c Id: 3/4

SIMULATION HARNESS – excluded from BOM/PCB

Scenario: battery-only → forced awake → USB hot-plug while BOOST_CMD remains asserted → autonomous charging → USB removal. Battery model includes 120 mΩ ESR. Battery model includes 120 mΩ ESR. I2 is the 3V3-converter input-power proxy. TP5613222A Input power/discharge is modeled Internally; legacy I3 is excluded from simulation.

